

Let's learn about **satellites!**

Space mission and **satellite** design course

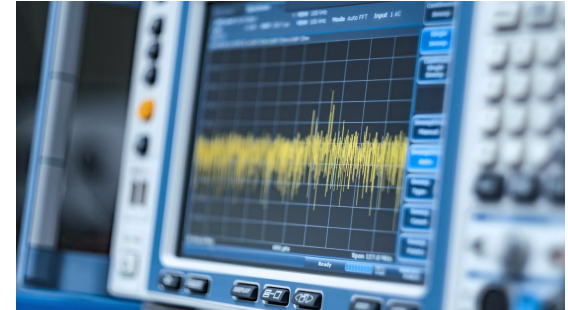
Payloads and Examples of Satellite Missions

Early Cubesat Missions

Early CubeSats were primarily designed for educational and technology demonstration purposes.

Initial missions aimed at demonstrating the feasibility of small-scale satellites in achieving specific objectives, given the platforms size constraints, limited power, and communication capabilities.

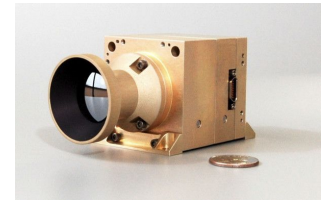
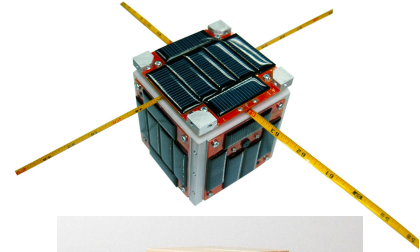
Objectives included fostering hands-on experience in space missions, testing new technologies, and conducting basic scientific experiments.



Early Cubesat Missions

Basic payloads of early CubeSat missions focused on Communication, Imaging, and Data Collection.

- **Communication:** Early CubeSats focused on establishing basic communication links, often using amateur radio frequencies. Limited bandwidth and data transfer rates were common.
- **Imaging:** Basic imaging systems were employed for Earth observation and educational purposes. Resolutions were modest compared to contemporary standards.
- **Data Collection:** Simple sensors were used for collecting basic scientific data. Payloads were constrained by the available space and power resources.



Challenges Faced with Early CubeSat Technology

The challenges of miniaturization, power limitations, and communication constraints directly impact CubeSat payloads.

Attempting to fit components into small dimensions poses engineering hurdles for reliable payload development. Power constraints affect payload functionality, and communication issues hamper data transmission.

Notably, statistics reveal that **out of 270 CubeSats launched between 2002 to 2016, 139 (51%) experienced mission failures**, often attributed to payload-related factors such as power failure and communication breakdowns.



These challenges highlight the intricacies in ensuring the success of CubeSat missions and the crucial role of well-designed payloads.

First missions

The first multiple launch CubeSat mission (Cubesat - Launch 1), involved 6 CubeSats as secondary payloads, and a primary commercial payload, took place June 30, 2003.

The launch vehicle was Rockot KS of Eurockot Launch Services (Bremen, Germany) from the launch site Plesetsk, Russia with the following secondary payloads:

- **XI** (X-factor Investigator) of the University of Tokyo, Japan
- **CUTE-I** of Tokyo Institute of Technology (TITech), Tokyo, Japan
- **CanX-1** of the University of Toronto, Toronto, Canada
- **AAUSat** of Aalborg University, Aalborg, Denmark
- **DTUSat** of the Technical University of Denmark, Lyngby, Denmark
- **QuakeSat** of Stanford University, Stanford, CA, USA (3U Cubesat with a total mass of about 4.5 kg)



XI-IV (X-factor investigator)

The goal of XI-IV was to validate a picosatellite bus system using commercial-off-the-shelf (COTS) parts, demonstrating successful on-orbit operation.

The satellite adheres to CubeSat program size and mass requirements, with 3-axis stabilization using a passive permanent magnet.

The XI-IV payload features a CMOS color camera for imaging experiments, with a detector size of 0.8 cm x 0.8 cm and 644 x 484 pixels in the visible spectrum.

The downlinked image contains 128 x 120 pixels due to limited data rate capacity.

The camera weighs 20 grams and consumes 120 mW of power.

As of December 26, 2020, the picosatellite and its instruments were confirmed to be operational.

Since October 2005, the picosatellite has been operational, although it faces two non-fatal issues: an A/D conversion anomaly caused by higher-than-expected solar cell output voltage and a recurrent on-board computer (OBC) reset due to a software error.



Parameter	Value	Parameter	Value
Size of satellite	10 cm x 10 cm x 10 cm	OBC Memory	PIC16LF877, 8bit, 4MHz
S/C mass	1 kg		EEPROM 256 kByte
Uplink	145 MHz FM, 1200 bit/s	Downlink (FM)	437 MHz FM, 1200 bit/s
Solar cells	Monolithic silicon (16%)	Antenna	437 MHz CW, 80 mW
Average power	1.1 W		Monopole (uplink)
Battery	Lithium-ion, 6.2 Ah		Dipole (downlink)
CW beacon	800 mW	S/C structure	Aluminum A7075
Thermal control	Passive	Attitude control	Passive with permanent magnet (1300mT), hysteresis damper

Table 1: Overview of spacecraft parameters

CUTE-I

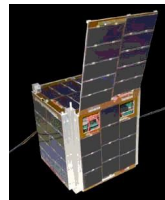
CUTE-I, a collaborative project between Tokyo Institute of Technology and Space Robotics and Teleoperations Laboratory, is a student-led CubeSat mission focusing on communication, sensing, and deployment tasks.

It employs a low-cost approach with COTS components. The satellite features solar-powered systems, including surface-mounted cells and deployable panels, and utilizes amateur radio standards for communication. Tasks included continuous telemetry transmission, sensor monitoring, and overseeing solar panel deployment.

CUTE-I has achieved mission success with a successful launch, deployment of antennas and solar cell paddle, and effective transmission of CW beacon and FM packets (AX.25 and SRLL) through uplinked commands. Downlink SRAM data and sensor tests were also successful.

The satellite initially experienced a stable spin until November 2003, attributed to slow rotation. Subsequently, instability arose, likely due to internal energy dissipation effects. Despite this, CUTE-I remains operational as of July 2004.

Communication subsystem



Communication subsystem	Tx1	CW transmitter output, 100 mW
	Tx2	FM transmitter output 350 mW
	Rx	FM receiver
	Antenna	Monopole x 3
		Ax.25 1200 bit/s (FSK)
Attitude sensors	TNC	Ax.25 1200 bit/s (FSK)
		MPU: H8/300
	CW Keyer	MPU: PIC
	Gyro (4)	4 axis
C&DH (Command and Data Handling)	Accelerometer	4 axis
	Sun sensor (CMOS type)	Original
	Thermistor	15 points
Power subsystem (generates voltage at 5.0 V, 3.7 V, and 3.3 V)	MPU	H8/300
	Memory	4Mbit SRAM
	Battery	Lithium-ion
	Solar cells	Silicon

CanX-1 (Canadian Advanced Nanospace eXperiment-1)

CanX-1 is the first picosatellite from University of Toronto. The mission aims to demonstrate a highly capable spacecraft with experimental subsystems, including testing CMOS imagers for Earth observation, moon observation, and star tracking, a powerful OBC based on an Atmel's ARM-7 microprocessor, a GPS receiver, and an active magnetic control system for detumbling and 3-axis stabilization.

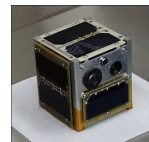
The power subsystem features lithium-ion batteries, solar arrays, a peak power tracker, and a power distribution module. The ADCS utilizes magnetic control with three torquer coils and a three-axis magnetometer.

The CanX-1 team faced challenges in establishing spacecraft communication. Despite sending ping frames and commanding a system reset during initial passes, no successful communication was achieved.

The ground team, however, could track and acquire beacons from other satellites in the mission, confirming the ground station's functionality.

Parameter	Color	Monochrome
Model	HDCS-20220	ADCS-2020
Quantum efficiency	33%	38%
Fill factor	42%	42%
Lens focal length	2.1 mm	25 mm
Lens aperture	f/2	f/2.5
Diagonal FOV	112°	14°
Nadir resolution	1.5 km/pixel (based on 650 km orbit altitude)	200 m/pixel
Power consumption	200 mW	200 mW

Table 3: Characteristics of the CMOS imagers



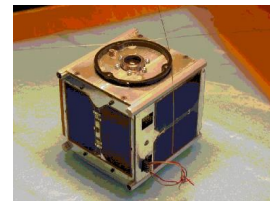
AAUSat (Aalborg University CubeSat)

AAU-CubeSat (or AAUSat) is a picosatellite developed by students at Aalborg University. The project involves multiple institutes at the university, focusing on five subsystems: ADCS, PSU, OBC, COM, and CAM (Imaging Camera).

The spacecraft is 3-axis stabilized, using magnetometers, sun sensors, and electromagnetic coils for attitude control. Power is supplied by triple-junction solar cells and lithium-iodine batteries. The OBC, featuring a 16-bit CPU from Siemens, handles data, controls ACS, and interfaces with other subsystems. RF communication is provided by an amateur radio link in the UHF-band (430-450 MHz) with a downlink data rate of 9.6 kbit/s using the AX.25 protocol.

The satellite is operated by students at Aalborg University, and images captured by the onboard Kodak MCM20027 camera chip are transmitted to the ground station and made available to the public. The imaging camera aims to capture scenes of Earth's surface, particularly Denmark, with a resolution of about 100 m x 100 m at an orbital altitude of 820 km.

After its launch on June 30, 2003, AAU-CubeSat (AAUSat) remained operational for two and a half months until the battery lost significant capacity. The satellite encountered challenges, with only limited data successfully downlinked due to an undisclosed problem affecting the transmitter, resulting in weak signals. Despite these issues, the project is deemed a success as a pioneering effort in picosatellite technology and student-built satellites.



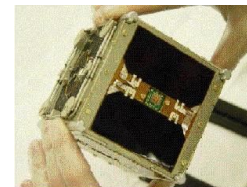
DTUSat

DTUSat, a student-defined and developed project at DTU (Technical University of Denmark), adheres to CubeSat specifications and features a monolithic wire-frame cube structure. The spacecraft has a three-axis stabilized configuration using magnetorquers for attitude control.

The sun angle sensing package, developed at DTU, employs MOEMS technology and consists of a two-axis linear slit sensor chip with a FOV of $\pm 70^\circ$. The power is provided by triple-junction solar cells, generating about 1.75 W. The OBC is based on an Atmel AT91M40800 processor, and RF communications use AMSAT concepts with uplink and downlink frequencies at 437.475 MHz.

DTUSat carries three payloads: **ETE** (Electromagnetic Tether Experiment), a **camera**, and a **radio test broadcaster**. ETE uses a bare copper wire for an electrodynamic tether experiment to change the spacecraft orbit without using fuel. The camera aims to capture Earth imagery with a CCD detector, and the radio test broadcaster supports spacecraft and AMSAT (radio amateur satellite) services.

After its launch on June 30, 2003, DTUSat faced communication issues, and no successful contact was established. Despite this setback, the project achieved its main goals of education for over 70 students, numerous outreach activities, and the development of a declared flightworthy satellite.

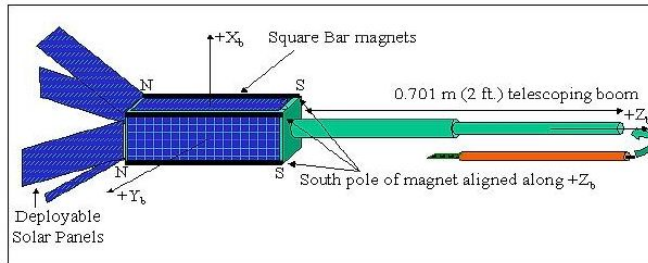


QuakeSat

QuakeSat is a research nanosatellite developed by the QuakeFinder Team of Palo Alto, CA, and SSDL (Space Systems Development Laboratory) at Stanford University. The primary objective of QuakeSat is to detect, record, and downlink Extremely Low Frequency (ELF) magnetic signals associated with earthquake activity. The nanosatellite, following the CubeSat concept, is a triple CubeSat configuration to accommodate a 30 cm long magnetometer on a telescoping boom.

The ADCS (Attitude Determination and Control Subsystem) uses permanent magnets to align the magnetometer close to the Earth's magnetic field. The EPS (Electric Power System) includes 12 solar arrays, Li-ion batteries, and power regulators. The C&DH (Command and Data Handling) system consists of a Prometheus CPU module with Ethernet and data acquisition, running a Linux operating system. RF communications use the UHF band with the AX.25 protocol.

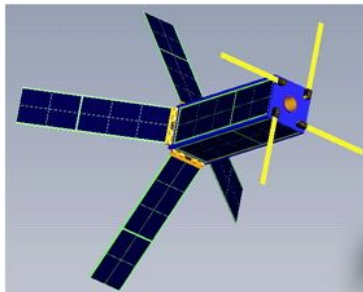
QuakeSat aims to investigate ELF magnetic waves associated with earthquake activity. The spacecraft is equipped with a magnetometer on a boom to measure these signals, which are believed to radiate from the earthquake hypocenter region through the Earth to the ionosphere. The nanosatellite, with a total mass of about 4.5 kg, exemplifies the CubeSat technology, utilizing Commercially-Off-The-Shelf (COTS) components for scientific experiments and payloads.



QuakeSat

QuakeSat, designed for a one-year mission, surpassed expectations by lasting 1.5 years until the end of December 2004. As of September 2004, the spacecraft and its instruments remained operational. The mission involved nearly 2000 magnetometer collections, primarily in mode 2 configuration (10 to 150 Hz), totaling around 500 MB of raw binary uncompressed data. Target areas included likely earthquake regions and post-earthquake collections. Additionally, over 100 collections were made to detect lightning strikes, and several global 1-10 Hz surveys were conducted.

By June 15, 2004, QuakeSat experienced periodic reboots due to brief eclipses, but it continued payload data collection and downloads. On January 20, 2004, after almost six months in low Earth orbit, both primary batteries failed within four days of each other, likely due to high temperatures reaching 120 degrees Fahrenheit. Despite this, QuakeSat operated on pure solar power, resulting in a significant degradation in mission performance, but it continued collecting and downloading nearly one-third of its previous data amount.



Modern CubeSat Missions

Modern CubeSat missions typically include:

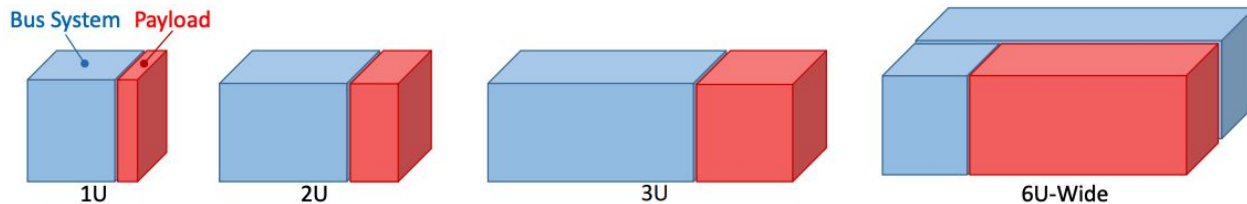
- Earth Observation
- Ionospheric Research
- Radiation Balance Measurement
- Space Weather
- Educational Missions
- Interstellar Travel Feasibility
- Solar Particles Observation
- Asteroid Prospecting
- Maritime Surveillance
- Space Weather Research



CubeSat Missions: Payload and Bus System

The mission's capability is influenced by the CubeSat's dimensions. The missions, key considerations include:

- **Optimized Bus Systems:** Larger CubeSats support advanced bus systems, enhancing mission capabilities.
- **Power and Storage Upgrade:** Increased size accommodates larger solar panels and batteries for elevated power and storage.
- **Advanced Processing and Communication:** Greater space enables powerful processors, expanded memory, and improved communication for efficient data handling.
- **Precise Attitude Control:** Integration of accurate attitude determination and control systems is facilitated in larger CubeSats.
- **Capacity for Larger Components:** Physical dimensions allow mounting larger scientific instruments, broadening experimental possibilities.



Payloads

The term "payload" in the context of space exploration, satellites, and spacecraft refers to the cargo or instruments that a vehicle is designed to carry and deliver to its destination.

It encompasses all the equipment, instruments, experiments, or other items that serve a specific purpose for the mission.

Payloads can vary widely depending on the goals and objectives of the mission and can include scientific instruments, communication devices, sensors, cameras, or any other equipment designed to perform tasks in space.

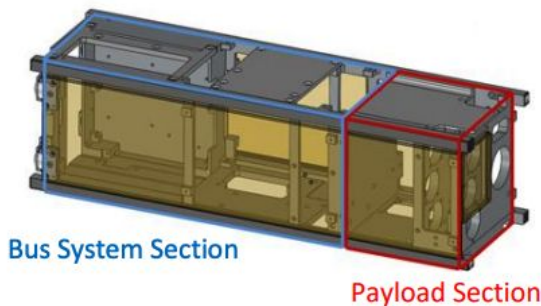


Payloads

In the case of CubeSats or small satellites, the payload is typically a crucial component as these satellites are often built for specific scientific, educational, or technological purposes.

Understanding the payload is essential for grasping the mission's objectives and the type of data or experiments the satellite is meant to conduct in space.

Therefore, the entire satellite mission is purposefully crafted around the payload.



Typical Payloads

Cubesats can host a diverse range of payloads:

- Optical Payloads: Optical and multispectral imaging cameras.
- Communication Payloads: Radio transponders, software-defined radios.
- Scientific Experiment Payloads: Instruments for space research.
- In-Orbit Demonstration (IOD) Payloads: Components or technologies validation.
- Navigation Payloads: Instruments for navigation experiments.
- Space Weather Payloads: Monitoring and studying space weather phenomena.
- Biological or Life Science Payloads: Experiments related to life sciences.
- Interplanetary Exploration Payloads: Instruments for missions beyond Earth.
- Astronomy Payloads: Instruments for observing celestial objects.
- Internet of Things (IoT) Payloads: Contributions to global IoT networks.
- Software Payloads: Experimental algorithms or software technologies.

Optical Payloads

Optical payloads for CubeSats encompass a variety of instruments designed to capture and analyze light, providing valuable data for a range of applications. There are different types of optical payloads commonly employed in CubeSats.

Imaging Payloads:

- Visible Cameras: CubeSats often carry visible light cameras for imaging Earth, celestial bodies, or specific regions of interest.
- Multispectral and Hyperspectral Cameras: These cameras capture images across multiple spectral bands, allowing for detailed analysis of environmental and Earth surface features.

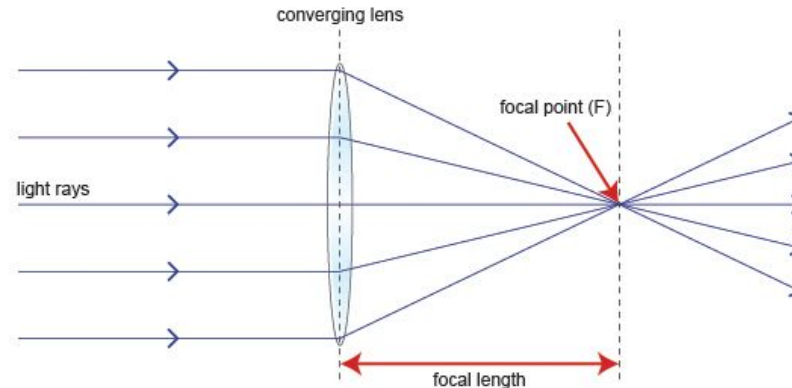


Visible Light Camera

A digital camera works by capturing and converting light into digital signals that can be processed and stored.



Light enters the camera through the lens, which focuses the light onto the camera's image sensor. The image sensor is a semiconductor device that consists of millions of tiny light-sensitive elements called pixels.



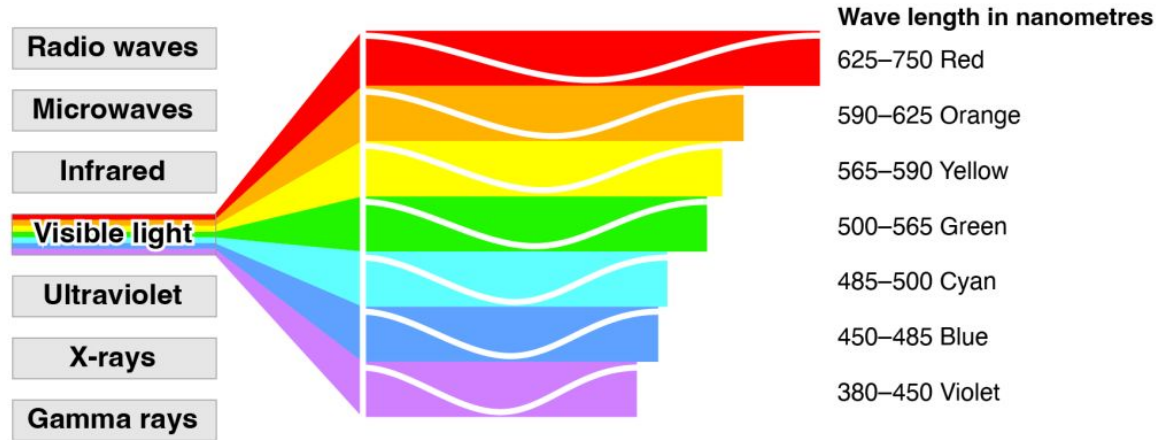
Visible Light Camera

Each pixel on the image sensor has a light-sensitive material (often a photosensitive diode or a photosite in a charge-coupled device, CCD, or a complementary metal-oxide-semiconductor, CMOS, sensor).



Visible Light Camera

When light strikes a pixel, it generates an electric charge in proportion to the intensity and color of the light.



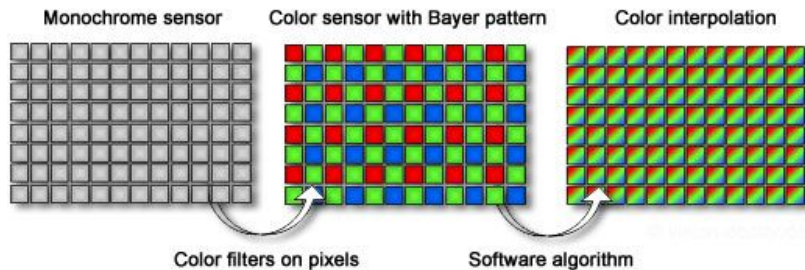
The accumulated electric charge in each pixel is then converted into a digital signal through a process called analog-to-digital conversion (ADC).

The ADC assigns a digital value to each pixel based on the amount of light it received.

Visible Light Camera

In color cameras, each pixel is typically overlaid with a color filter, allowing it to capture information about the intensity of red, green, and blue light.

The camera combines the information from neighboring pixels to reconstruct the full color of each point in the image.



Visible Light Camera

The digital signals from the image sensor are processed by the camera's onboard electronics, including a Digital Signal Processor (DSP).

The DSP performs various tasks such as white balance adjustment, noise reduction, and sharpening to enhance the quality of the image, via appropriate filters.

Filters serve multiple purposes:

- Noise reduction: Filtering techniques, such as Gaussian smoothing or median filtering, remove random variations and disturbances in pixel values caused by sensor imperfections or transmission errors.

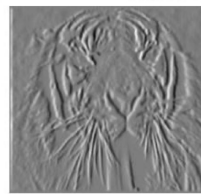


Visible Light Camera

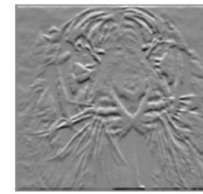
- Edge detection: Filters like the Sobel or Laplacian filter emphasize rapid intensity changes (edges and boundaries), aiding in the identification of important features and structures.



Image f

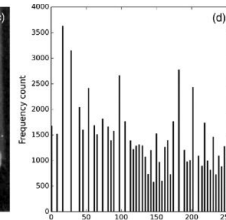
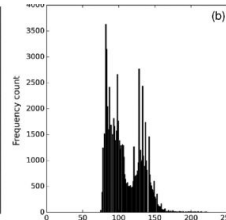


$$\frac{\partial f(x, y)}{\partial x}$$



$$\frac{\partial^2 f(x, y)}{\partial x^2 + \partial y^2}$$

- Enhancement of image features: Techniques like contrast stretching, histogram equalization, or the application of specific frequency filters help bring out certain features.



Visible Light Camera

- Frequency analysis: Transformations like the Fourier Transform allow the analysis of an image's frequency components, helping to identify patterns, textures, and structures.



- Image compression: Filters may be used to identify and retain essential image information, minimizing redundant or less critical data during compression.

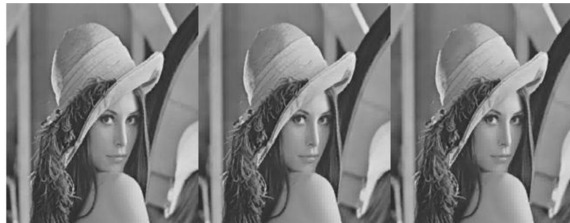


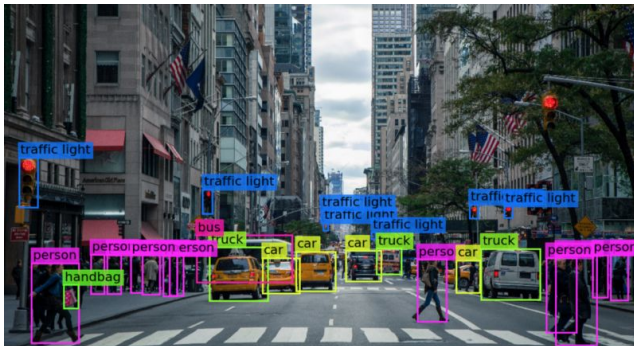
Figure 2: Reconstructed image of previous method and present work for HP values=10,20,30,40,50,60 shown from top left

Image Filtering

Filters play a crucial role in computer vision by enhancing or modifying the visual information contained in images.

Computer vision is a multidisciplinary field of AI that empowers machines to interpret and make decisions based on visual data. It aims to enable computers to mimic human vision by recognizing patterns, extracting meaningful information, and understanding the content of images or videos.

Computer vision encompasses a wide range of tasks, including image recognition, object detection, segmentation, facial recognition, and scene understanding.



Computer Vision: An Example

Let's have a look at the following image of LEGO bricks:



How do we create an algorithm that is capable of recognizing, counting and classifying the bricks from a picture taken by the camera?

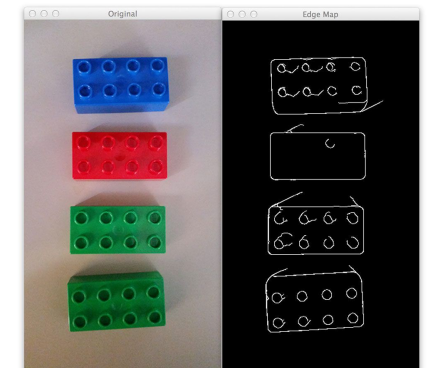
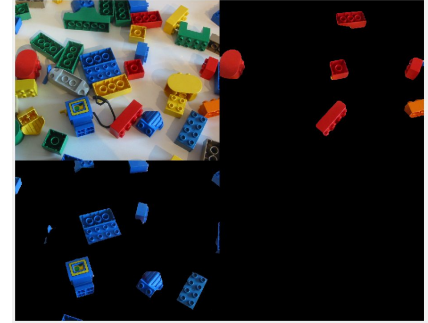
Which features would you look for to do so?

Computer Vision: An Example

Recognizing, counting, and classifying a set of LEGO bricks in an image involves several steps.

A simplified example using common computer vision techniques:

- **Image Acquisition:** Begin by obtaining an image containing LEGOs. This image can be captured using a camera or sourced from an existing dataset.
- **Preprocessing:** Apply preprocessing techniques to enhance the quality of the image. This may include resizing, normalization of pixel values, and noise reduction using filters.
- **Color Segmentation:** LEGO bricks often have distinct colors. Apply color segmentation to separate the image into regions corresponding to different colors. This can be achieved by thresholding or clustering algorithms.
- **Edge Detection:** Use edge detection filters, such as the Canny edge detector, to identify the boundaries of individual LEGO bricks. This step helps in separating adjacent blocks and improving accuracy during subsequent processing.



Computer Vision: An Example

- **Contour Detection:** Identify contours of the segmented regions. Each contour corresponds to a potential LEGO brick.
- **Size and Shape Filtering:** Filter out contours based on size and shape criteria to eliminate small artifacts or non-brick elements. This step helps focus on relevant regions that are likely LEGOs.
- **Counting:** Count the number of remaining contours after filtering. Each contour corresponds to a LEGO brick.
- **Feature Extraction:** Extract features from each remaining contour, such as color, texture, or geometric properties. These features will be used for classification.
- **Classification:** Train a machine learning model, such as a simple classifier or a more sophisticated neural network, using the extracted features. The model should be trained to classify LEGOs into different types based on the features observed.
- **Classifying Blocks:** Apply the trained model to classify each remaining contour as a specific type of LEGOs (e.g., brick, plate, or tile). This step enables the algorithm to recognize and label each block in the image.

Computer Vision and Camera Payload

CubeSat's camera payload can integrate a dedicated processor, enabling real-time image manipulation and analysis through AI and neural networks.

This self-contained system allows for onboard processing, reducing the need for immediate data transmission to the ground station, removing unnecessary or heavy processing load from the onboard computer (OBC) and enhancing operational efficiency during Earth observation missions.



Multispectral & Hyperspectral Cameras

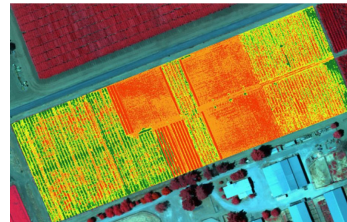
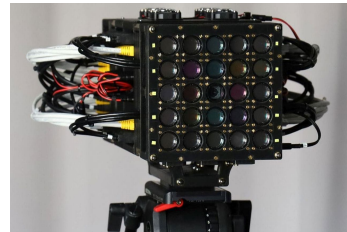
Multispectral cameras are designed to capture information in multiple discrete bands or specific wavelength ranges.

In contrast to standard RGB imaging, which utilizes three bands (red, green, and blue), multispectral cameras typically capture information in a few distinct bands, often ranging from 3 to 10.

These cameras find applications in various fields, including agriculture, environmental monitoring, and remote sensing. They are particularly valuable in tasks such as vegetation analysis, land use classification, and resource management.

One characteristic of multispectral cameras is their moderate to high spatial resolution.

The higher the spatial resolution the higher the ability to capture finer details in an image.



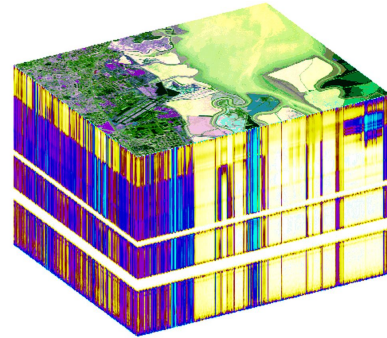
Multispectral & Hyperspectral Cameras

Hyperspectral cameras, on the other hand, capture information across a continuous range of narrow and contiguous spectral bands.

Unlike multispectral cameras, hyperspectral devices are capable of capturing data in hundreds of bands, offering fine spectral resolution.

The advanced capabilities of hyperspectral cameras are well-suited for specialized applications, including environmental monitoring, mineralogy, and defense. Their extensive spectral information allows for the identification of subtle variations in material composition.

However, the processing of hyperspectral data is more complex, often involving advanced techniques such as spectral unmixing and classification algorithms.



Optical Payloads

Space Science Optical Payloads:

- Telescopes: CubeSat telescopes are used for astronomical observations, studying stars, galaxies, and other celestial objects.
- Spectrometers: These instruments analyze the spectrum of light emitted or absorbed by objects, providing insights into their composition and properties.



Optical Payloads

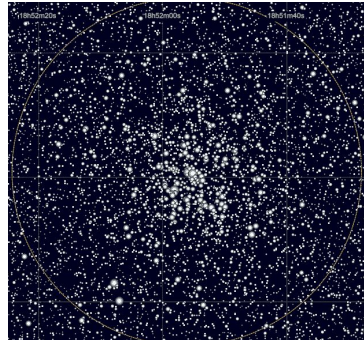
Star Trackers:

A star tracker functions as an optical navigation device in spacecraft, determining their orientation in space by capturing images of the night sky and precisely identifying the positions of stars.

Equipped with a predefined star catalog, the tracker's optical system detects and records star patterns within its limited field of view.

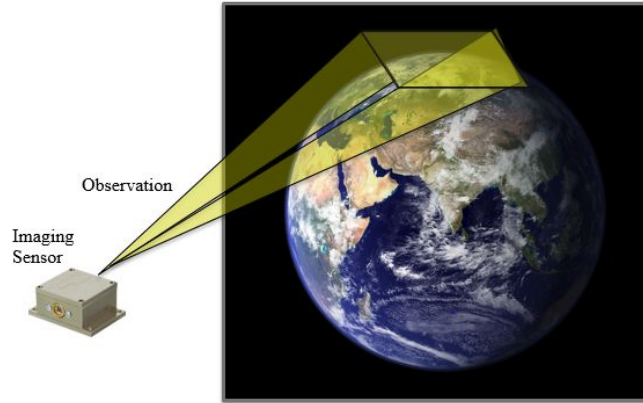
An onboard computer processes these images, employing sophisticated algorithms to match observed stars with their known positions in the catalog. Through triangulation, the tracker calculates the spacecraft's three-dimensional attitude.

Continuous updates are made as the tracker captures new images and corrects for changes in star positions over time.



Optical Payloads

Earth Horizon Sensors: These sensors determine the spacecraft's position relative to the Earth's horizon, aiding in attitude control.



Optical Payloads

Lidar (Light Detection and Ranging):

CubeSats may utilize lidar systems for remote sensing applications, measuring distances by sending laser pulses and analyzing the reflected light.



Optical Payloads

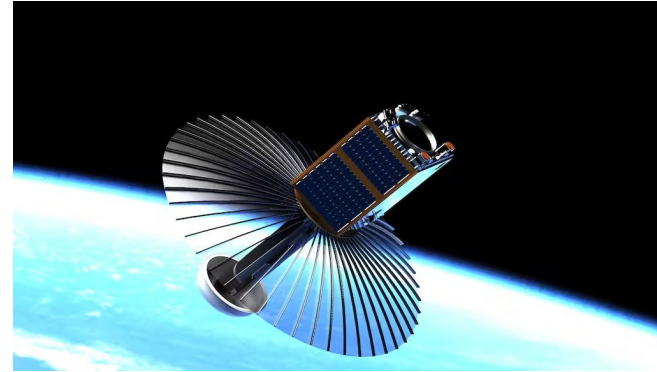
Synthetic Aperture Radar (SAR):

SAR is a sophisticated remote sensing technology that utilizes microwave signals emitted from a sensor platform pointed towards the Earth's surface.

By synthesizing a virtual aperture through the motion of the sensor platform, SAR processes the recorded signals to achieve superior range and azimuth resolution.

High-resolution SAR images that provide valuable insights into topography, vegetation, and environmental changes.

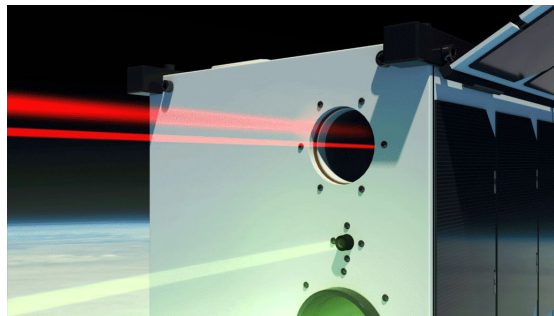
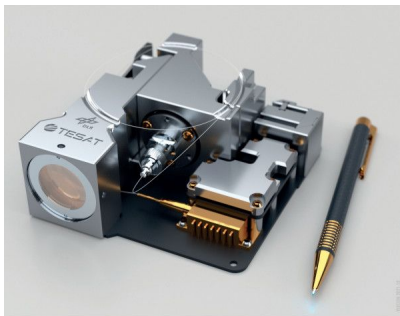
SAR stands out for its all-weather capabilities and the ability to produce detailed images independent of daylight.



Optical Payloads

Optical Communication Systems:

CubeSats can carry optical communication payloads for high-speed data transmission using lasers. This technology offers advantages over traditional radio-frequency communication (we will see later in the slides).



Communication Payloads

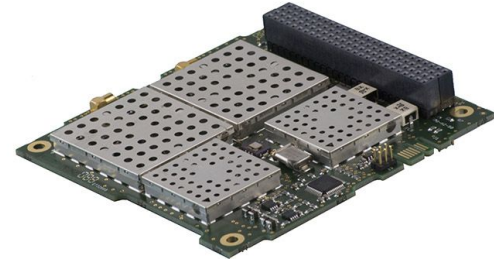
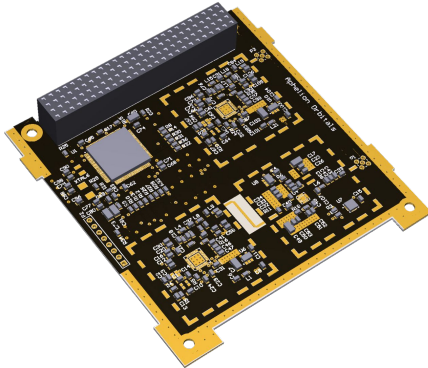
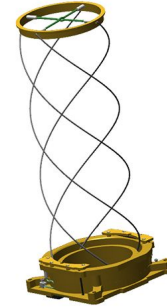
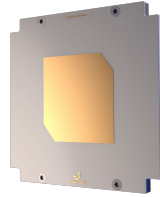
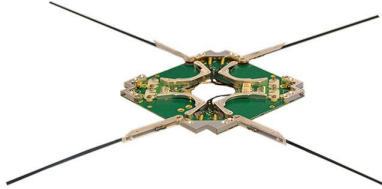
Common types of communication payloads found on cubesats:

- **UHF/VHF Transceivers (300 MHz to 3 GHz):** Ultra High Frequency (UHF) and Very High Frequency (VHF) transceivers are commonly used for communication with ground stations. They operate in the amateur radio bands and allow cubesats to establish two-way communication.
- **S-Band Transceivers (30 MHz to 300 MHz):** S-Band transceivers operate in the microwave frequency range and are used for higher data rates. They provide better communication capabilities compared to UHF/VHF, making them suitable for missions with higher data transfer requirements.
- **X-Band Transceivers:** X-Band is another microwave frequency band used for communication in cubesats. It offers higher data rates than S-Band and is suitable for missions that demand significant amounts of data to be transmitted.
- **Software-Defined Radios (SDRs):** SDRs provide flexibility in terms of communication frequencies and modulation schemes. They allow for reprogramming the communication system onboard, enabling adaptation to different mission requirements.
- **Antennas:** Various antenna configurations are used depending on the mission and frequency band. Deployable antennas, such as tape or whip antennas, can be used to increase the effective communication range.

Communication Payloads

- **Store-and-Forward Systems:** Some cubesats use store-and-forward systems, where they collect data and store it onboard until they pass over a ground station. At that point, they transmit the stored data to Earth.
- **Inter-Satellite Communication:** In some cases, cubesats are equipped with communication systems to establish links with other cubesats or satellites. This is particularly relevant for constellations of satellites working together.
- **Amateur Radio Payloads:** Many educational cubesats carry amateur radio payloads that allow students and amateur radio operators to communicate with the satellite. This serves educational purposes and encourages public participation.
- **Optical Communication (Laser Communication):** Advanced cubesats may experiment with optical communication systems, using lasers to transmit data between the satellite and ground stations. This technology offers higher data rates compared to traditional radio frequencies.

Communication Payloads

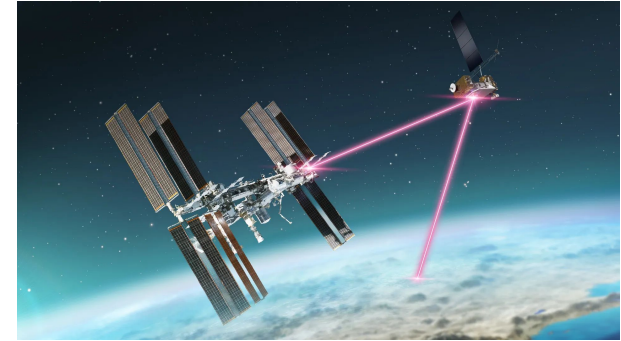


Optical Communication

Optical communication, also known as laser communication or free-space optical communication (FSO), for CubeSats involves the use of lasers to transmit data through free space, typically in the form of laser beams. This method offers advantages such as high data transfer rates, low power consumption, and potential for small form factors.

Components involved:

- **Laser Transmitter:** The CubeSat is equipped with a laser transmitter that emits laser beams to carry data. The transmitter may use modulated lasers to encode information in the form of light pulses.
- **Telescope or Optics:** A telescope or optics system on the CubeSat is used to focus the laser beam and enhance the precision of communication. The optics help maintain a directed and collimated beam for efficient transmission.
- **Communication Protocol:** Various communication protocols, such as On-Off Keying (OOK) or Pulse Position Modulation (PPM), may be employed to modulate the laser signals and encode data for transmission.
- **Laser Safety:** Safety measures are crucial to prevent harm to other satellites or terrestrial objects. CubeSat laser communication systems must comply with international regulations to minimize the risk of interference and accidental exposure.



Optical Communication

Advantages of optical communication:

- **High Data Rates:** Optical communication can provide high data transfer rates compared to traditional radio frequency (RF) communication.
- **Low Power Consumption:** Laser communication systems often have lower power requirements than RF systems, making them suitable for CubeSats with limited power budgets.
- **Small Form Factor:** Optical components can be more compact than their RF counterparts, allowing CubeSats to integrate laser communication systems within their small form factors.
- **Secure Communication:** Laser communication is more directional and can be harder to intercept compared to RF signals, providing a level of security for data transmission.
- **Reduced Frequency Congestion:** Optical communication operates in a different part of the electromagnetic spectrum, reducing the likelihood of interference or congestion in commonly used RF frequency bands.

Disadvantages of optical communication:

- **Line-of-Sight Requirement:** Optical communication relies on a clear line of sight between the transmitting and receiving systems. This can be challenging in the dynamic orbital environment.
- **Atmospheric Effects:** Atmospheric conditions can affect the performance of optical communication, especially during the satellite's passage through Earth's atmosphere.
- **Tracking and Pointing:** Accurate tracking and pointing systems are essential to maintain a stable connection between communicating satellites. This is crucial for CubeSats with limited resources for sophisticated pointing systems.
- **Regulatory Compliance:** CubeSat missions using optical communication must comply with international regulations and standards to avoid interference with other space-based and terrestrial systems.

Scientific Experiment Payloads

A CubeSat can be outfitted with specialized instruments or sensors designed to conduct scientific observations, measurements, or experiments in space.

The primary purpose of these payloads is to collect data related to the scientific goals. This could involve capturing images, measuring environmental parameters, detecting certain particles or radiation, or conducting experiments to study phenomena in space.

The scientific payloads are integrated into the CubeSat's structure, ensuring compatibility with the satellite's overall design. Integration involves considerations such as power supply, communication interfaces, and physical accommodation within the CubeSat's form factor.

CubeSats typically rely on solar panels for power. The scientific payloads must be designed to operate within the power budget of the CubeSat, ensuring that energy requirements are met and that the satellite can sustain the scientific experiments throughout its mission.

The CubeSat must be equipped with communication systems to transmit data collected by the scientific payloads back to Earth. This involves establishing telemetry links and using suitable communication protocols.

Scientific payloads need to withstand the harsh conditions of space, including temperature extremes, radiation, and the mechanical stresses during launch and deployment.



Scientific Experiment Payloads

Scientific experiment CubeSats examples:

- **Magnetometer Payloads:** CubeMag aimed at measuring the Earth's magnetic field, studying space weather, and testing miniaturized magnetometer technology.
- **Radiation Detection Payloads:** RadFXSat carried a space weather experiment to study the effects of radiation on commercial off-the-shelf (COTS) electronic components.
- **Biology Experiments:** BioSentinel planned to carry yeast cells to study the impact of deep space radiation on living organisms.
- **Space Weather Monitoring:** IRVINE02 studied the ionospheric response to lightning-generated Very Low Frequency (VLF) and Low-Frequency (LF) radio waves.
- **Plasma Experiment:** QB50 consisted of multiple CubeSats equipped with Langmuir probes to study the Earth's thermosphere and ionosphere.
- **Astronomy Payloads:** Aalto-1 carried a compact radiometer to study the Earth's thermal radiation in the infrared spectrum.
- **Technology Demonstrations (IOD):** TechEdSat included experiments to test deployable structures, a deorbit sail, and a smartphone-based satellite control system.

Brokkr-1

AstroForge, a startup leading the way in asteroid mining, is all set for two big missions. The first one, Brokkr-1, is a 6U cubesat launched on SpaceX's Transporter 7 in 2023.

Brokkr-1 Highlights:

- Built by OrbAstro, Brokkr-1 it's IOD represents a crucial event for AstroForge's future mission on larger satellites.
- Brokkr-1 is designed to experiment with and validate the technology for extracting metals from materials that simulate the composition of asteroids.



GreenCube

GreenCube is a 3U CubeSat designed to showcase an autonomous biological laboratory for plant cultivation on a CubeSat platform.

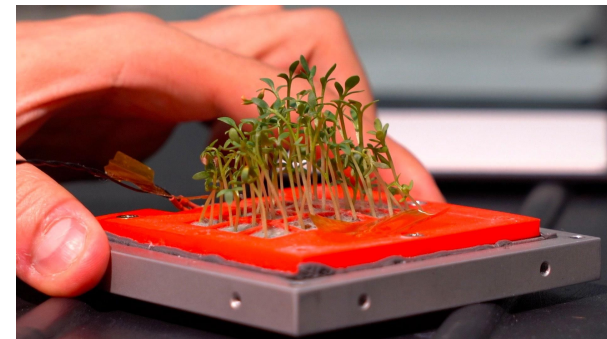
Key Features:

- Launched aboard the VEGA-C rocket by the European Space Agency.
- Micro-garden measures 30 x 10 x 10 centimeters.
- Closed-loop hydroponics with advanced environmental control systems.
- Provides a 20-day growth cycle for selected microgreens.
- Housed in a pressurized environment with hi-tech remote monitoring sensors.



GreenCube
Microgreens cultivation
in a CubeSat

A project coordinated by the Italian Space Agency



GreenCube

Mission Objectives and Significance:

- GreenCube aims to contribute to bio-regenerative systems for life support in space.
- Microgreens crucial for fresh food, supplementing astronaut diets, which now are provided with pre-packaged meals.
- Psychological benefits through connection to Earth's environment to astronauts and reduce stress associated with isolation in space.

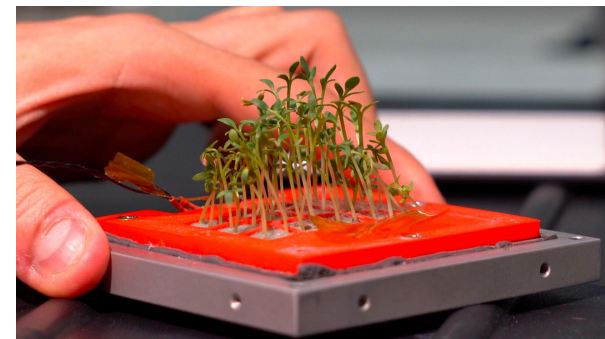
Payload:

- Micro-garden and sensor units.



GreenCube
Microgreens cultivation
in a CubeSat

A project coordinated by the Italian Space Agency



In-Orbit Demonstration (IOD) of Subsystems

In the realm of CubeSats, the concept of In-Orbit Demonstration (IOD) emerges from the flexibility that any subsystem within a CubeSat can act as a payload.

This means that not only is the overall mission payload-centric, but individual components or experiments within the CubeSat – be it a new communication module, a novel sensor, or an advanced power system – can themselves serve as payloads for validation in space.

This dual role emphasizes the versatility of CubeSats and underscores the significance of IOD missions in showcasing and validating innovations on a smaller scale before broader implementation.

E.g. a 3U satellite having a payload OBC in addition to the main OBC.

In-Orbit Demonstration (IOD) of Subsystems

While the "unknown" category poses a significant source of errors in the initial stages, subsystems play a crucial role in contributing to CubeSat failures. Beyond the first 30 days, the Electrical Power System (EPS) emerges as a primary cause, followed by the Communication subsystem after 90 days. This data underscores the vital importance of In-Orbit Demonstration (IOD) missions.

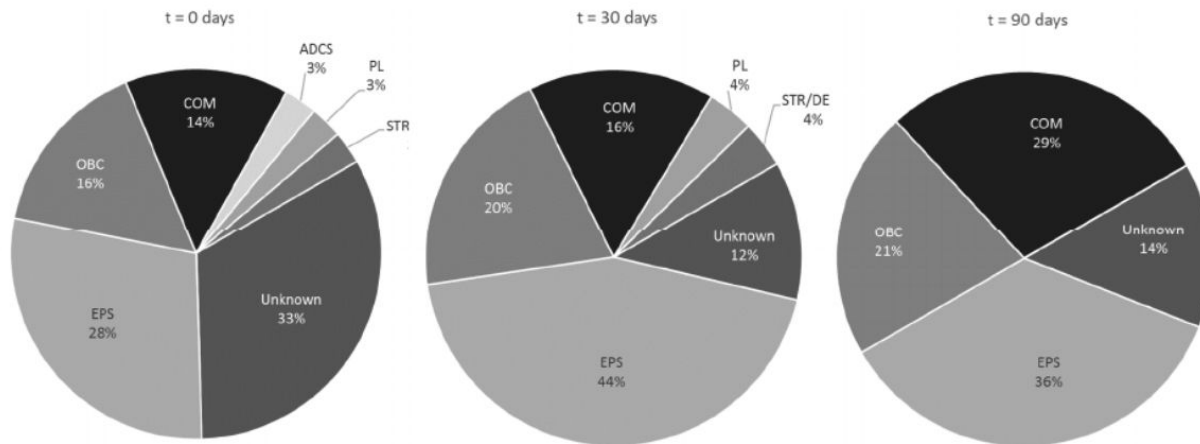


Figure 5: Subsystem contributions to CubeSat failure after ejection (incl. DOA), 30 days and 90 days

Interplanetary Exploration Payloads

Interplanetary exploration payloads on CubeSats are designed to extend the capabilities of these small satellites beyond Earth's orbit, enabling scientific investigations and data collection in our solar system.

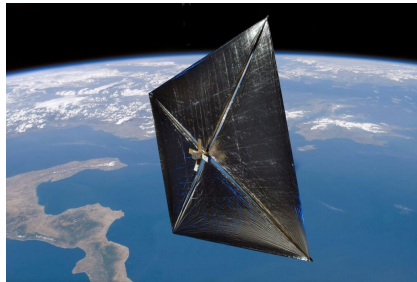
While traditional interplanetary missions involve larger spacecraft, CubeSats (sometimes) offer a cost-effective and flexible platform for exploring nearby celestial bodies.

Interplanetary exploration payloads involve:

- Robust communication systems to relay data back to Earth from large distances. This involves advanced radio systems with higher power and specialized antennas.
- Miniature propulsion systems for trajectory adjustments, orbit changes, or even interplanetary transfer. These systems can include miniaturized thrusters or innovative propulsion technologies like solar sails.
- Constellations: Instead of a single CubeSat, missions may involve constellations of these small satellites working together to gather data from multiple vantage points.

Interplanetary Exploration Payloads

- Entry, Descent, and Landing Systems: For missions targeting bodies with atmospheres, CubeSats may include systems for entry, descent, and landing. This can involve heat shields, parachutes, or other innovative technologies for safely reaching the surface.
- Interplanetary CubeSats require autonomous navigation systems to make decisions based on real-time data.
- Deployable Sensors: Some interplanetary CubeSats may carry deployable sensors or instruments, such as landers or probes that detach from the main satellite to study the surface of a celestial body up close.



MarCO

The MarCO mission aimed to revolutionize interplanetary communication by providing real-time support during the NASA's InSight Lander's descent onto Mars.

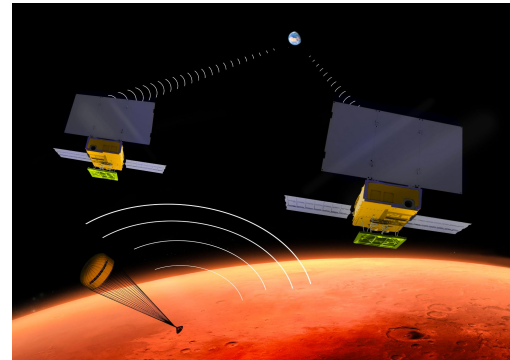
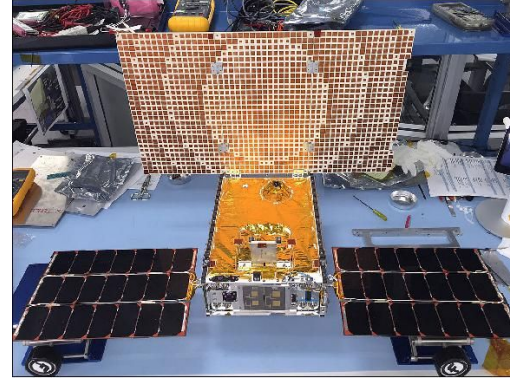
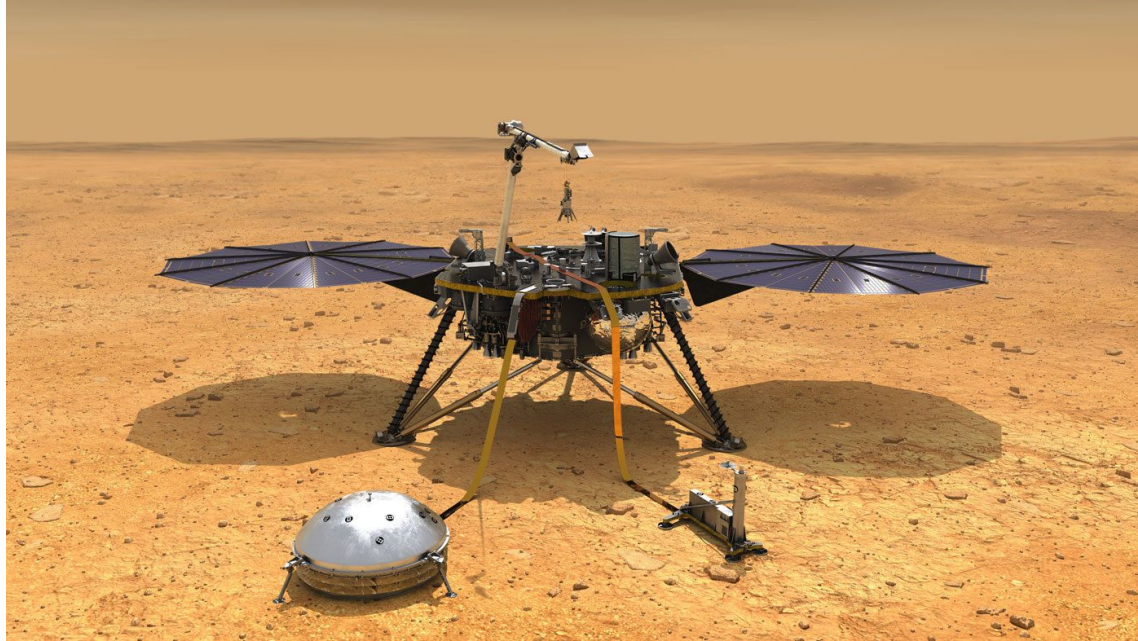
Key Aspects:

- Two 6U Cubesats, MarCO-A and MarCO-B, served as crucial components in relaying essential data from the InSight lander during its descent onto the Martian surface.
- The mission focused on establishing an immediate and reliable communications link between Mars and Earth, allowing for the real-time monitoring of InSight's landing.

Relevance of Experiment:

- The successful execution of real-time communication during the descent phase significantly enhanced our understanding of the Martian environment.
- This experiment showcased the potential of 6U CubeSats for rapid and vital data transmission in deep space, marking a milestone for future interplanetary missions.

MarCO



Internet of Things (IoT) Payloads

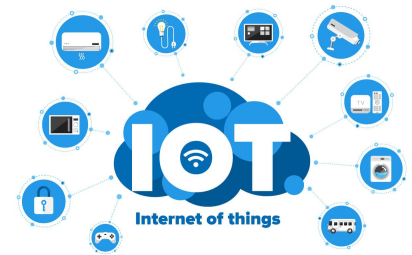
The Internet of Things (IoT) revolutionizes connectivity by linking physical devices, objects, and systems through embedded sensors, actuators, and network connectivity. IoT devices collect data on various parameters, such as temperature and motion, using sensors, and communicate this data over the internet for analysis and action.

In the IoT ecosystem, devices often have embedded processors for local data processing, while cloud platforms handle centralized data storage and analysis. Security measures, including encryption and authentication, safeguard the sensitive data transmitted by IoT devices.

Automation is a core feature, allowing devices to make autonomous decisions based on predefined conditions without direct human intervention.

Interoperability is critical for seamless collaboration among devices, requiring standardized communication protocols. IoT applications span diverse domains, including smart homes, healthcare, agriculture, and industrial automation, improving efficiency and decision-making.

Big data analytics plays a key role in extracting valuable insights from the vast data generated by IoT devices. As technology advances, IoT continues to grow, creating a more connected and intelligent world that enhances daily life and transforms various industries.

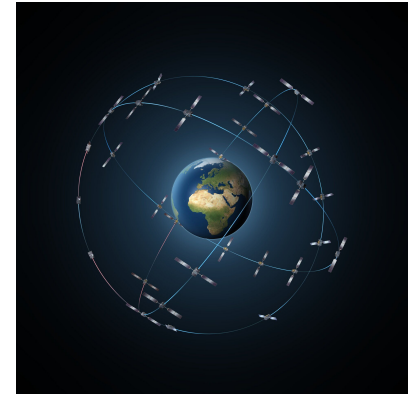


Internet of Things (IoT) Payloads

In the quest for enhanced global connectivity, the fusion of IoT with space technologies takes a significant leap forward as specialized payloads find their place on CubeSats. These can incorporate IoT capabilities designed for seamless uplink and downlink communication for real-time data collection and transmission.

Global coverage is achieved through constellations of CubeSats in Low Earth Orbit (LEO), providing comprehensive monitoring from different geographic locations. Orbit considerations are tailored to the specific IoT application, and CubeSats act as remote sensors, collecting data from remote or inaccessible areas. Data fusion algorithms enhance analysis by integrating information from various sensors. CubeSats may deploy IoT sensor nodes or buoys to enhance spatial coverage and enable targeted data collection.

These IoT payloads can be customized for applications such as agriculture, environmental monitoring, disaster response, and wildlife tracking. Inter-satellite communication capabilities in CubeSat networks foster collaboration and data sharing, amplifying the overall effectiveness of the IoT network. The flexibility of CubeSats allows for a broad range of tailored IoT missions.

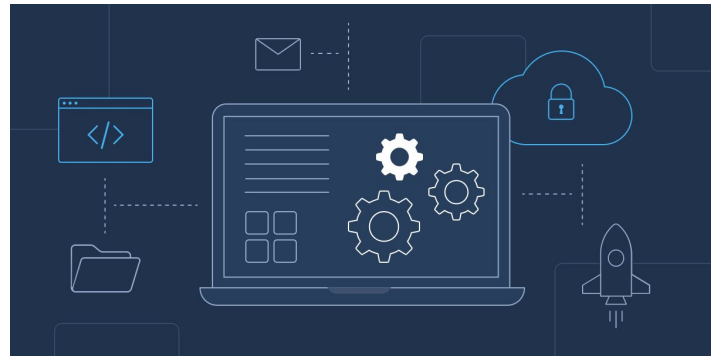


Software Payloads

While traditional payloads often involve physical instruments or sensors, software payloads focus on the implementation of specific algorithms, data processing, and control systems.

Software payloads can be:

- Data Processing Algorithms
- Autonomous Control Systems
- Communication Protocols
- Experiment Control and Sequencing
- Machine Learning Applications
- Security Systems
- Onboard Software Updates

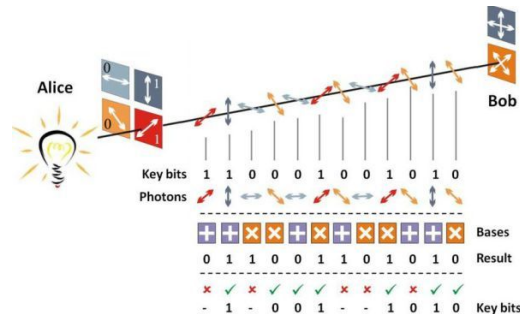


Advanced technologies

Thanks to the adaptability and potential of CubeSats, several innovative technologies are currently being developed for testing on these platforms:

- **Quantum Communication Experiments:** CubeSats conducting quantum key distribution experiments use entangled photons for secure communication, showcasing advancements in quantum information science within a small satellite form factor.

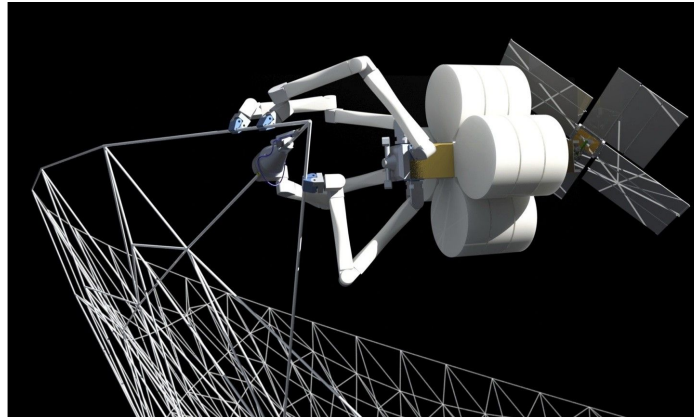
Quantum Key Distribution (QKD) is a cutting-edge technology that exploits quantum mechanics to securely exchange cryptographic keys between parties, typically Alice and Bob. In QKD, quantum properties like superposition and entanglement are used to encode and transmit information in the form of quantum bits (qubits). The entangled nature of particles ensures that any eavesdropping attempts would disturb the quantum states, alerting the communicating parties. By comparing measurements through a classical channel, Alice and Bob can detect and discard compromised bits, ultimately creating a secure shared key. QKD provides a fundamentally secure method for communication, resistant to potential eavesdropping.



Advanced technologies

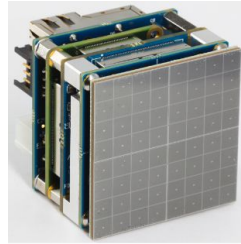
- **In-Orbit Manufacturing Technology:** CubeSats experimenting with in-orbit manufacturing technologies, such as 3D printing in microgravity, push the boundaries of space manufacturing and resource utilization.

The significance lies in the potential for on-the-spot production of satellite components or structures in space, eliminating the need for Earth-based manufacturing and the associated challenges of transporting pre-built components. This not only reduces the complexity and cost of satellite launches but also opens up possibilities for adapting and customizing satellite designs in response to evolving mission requirements.

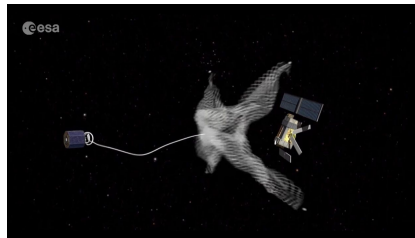


Advanced technologies

- **Miniaturized X-ray or Gamma-Ray Detectors:** CubeSats with miniaturized detectors for X-rays or gamma rays contribute to space-based astrophysics and are innovative in enabling compact yet powerful instruments for celestial observations.



- **Space Debris Removal Technologies:** CubeSats testing technologies for capturing or deorbiting space debris contribute to the development of innovative solutions for mitigating the growing issue of orbital debris in Earth's vicinity.



Advanced technologies

Solar Sail Propulsion Systems: CubeSats equipped with deployable solar sails showcase innovative propulsion technology that harnesses the pressure of sunlight for efficient and continuous acceleration.

Solar sails for CubeSats operate based on the principles of light propulsion. Unlike traditional chemical propulsion systems, solar sails utilize the pressure of sunlight to generate thrust, enabling CubeSats to maneuver and travel in space. How solar sails work:

- **Photon Pressure:** Solar sails rely on the pressure of photons, which are particles of light, to create thrust. While photons do not have mass, they carry momentum, and when they reflect off a surface, they transfer this momentum to the object.
- **Thin, Reflective Material:** The solar sail is typically made of an ultra-thin and reflective material, such as Mylar or other lightweight polymers. This material is stretched across a framework to form a large, thin sail.
- **Acceleration and Maneuvering:** The acceleration generated by the solar sail allows CubeSats to maneuver in space. By adjusting the orientation of the sail relative to the incoming sunlight, CubeSat operators can control the direction of thrust and thus change the satellite's trajectory.
- **Continuous Acceleration:** While the thrust from a solar sail is modest, it is continuous and can accumulate over time. This allows CubeSats equipped with solar sails to achieve significant velocity changes and travel to destinations within the solar system.

Advanced technologies

Innovative Power Harvesting: CubeSats experimenting with innovative power harvesting methods, such as deployable solar concentrators or energy scavenging from the space environment, demonstrate advancements in sustainable energy solutions.

Electrodynamic tethers can be employed for energy harvesting in addition to propulsion. In the context of energy harvesting, the tether interacts with the Earth's magnetic field to generate electrical power rather than propulsion. The basic principle involves moving a conductive tether through the Earth's magnetic field, inducing an electromotive force (EMF) in the tether.

As the tether cuts through the Earth's magnetic field lines, the relative motion induces a voltage along the tether. By connecting the ends of the tether through an electrical load, a current is created, and electrical power is generated. This process allows for the conversion of kinetic energy from the spacecraft's orbital motion into electrical energy.

This concept of using electrodynamic tethers for energy harvesting is known as a "rotating tether," and it represents a potential method for generating power in space. It offers a sustainable and efficient way to produce electrical energy for spacecraft, especially those in orbits where traditional solar power generation may be challenging, such as in regions with extended periods of darkness.



Thanks for the attention!